Live Human Activity Recognition using Smartphone Sensors

Recognising what a person is doing from phone motion data, live in the browser

T E A M

K Srujana - PES1UG24AM128

Moksha S - PES1UG24AM167

The Problem

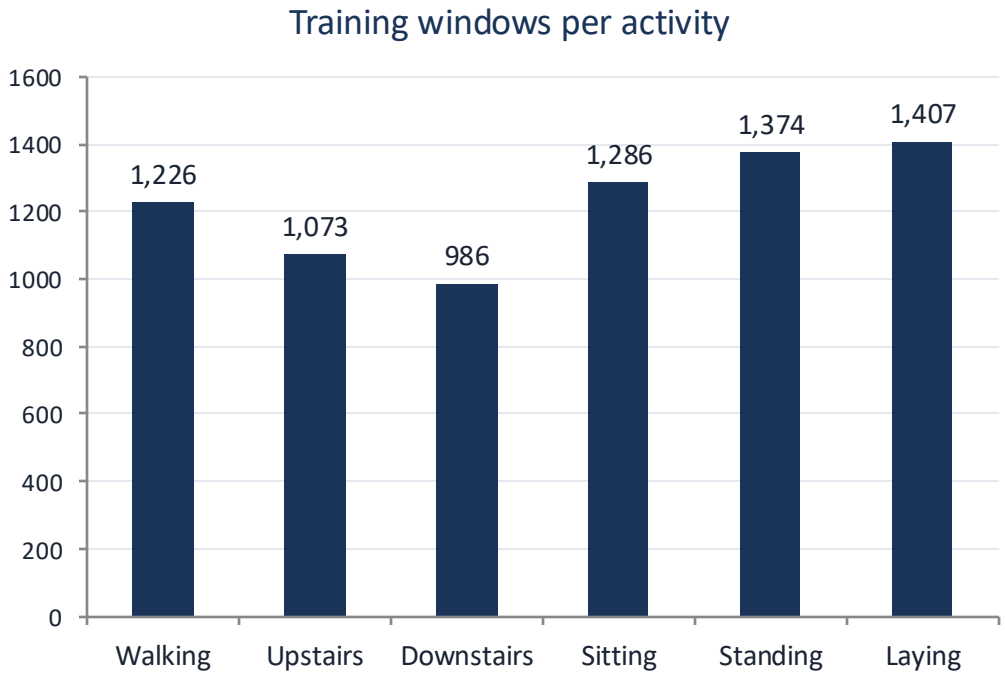
Classify every 2.56 s window of motion into 1 of 6 activities

- Walking
- Walking upstairs
- Walking downstairs
- Sitting
- Standing
- Laying

Why it matters

- Phones already carry an accelerometer and gyroscope, so no extra hardware
- Useful for fitness tracking and elderly-care monitoring
- Context-aware apps can react to what you are doing
- Our goal: make it work live in a phone browser

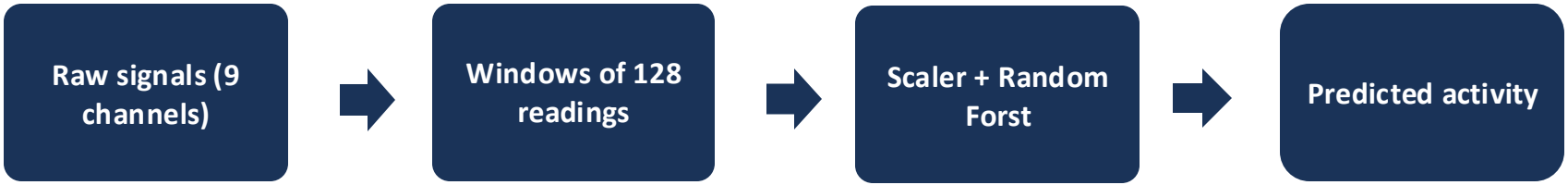
Dataset: UCI HAR



10,299	7,352	2,947
windows in total	train	test

- 30 volunteers (19 to 48 years) with a phone at the waist
- Accelerometer + gyroscope at 50 Hz, labelled from video
- Windows of 128 readings with 50% overlap
- Train and test are different people
- Fairly balanced classes

Approach & Pipeline



Model: Random Forest

- StandardScaler, then 250 trees
- Balanced class weights, fixed seed
- Fast, robust, small file (about 15 MB)

Live app: Streamlit

- Phone browser reads the motion sensors
- Same 96-feature code as training
- Vote over the last 5 predictions
- Needs HTTPS (Streamlit Cloud)

Results & Verification

96	87.82%	6
features per window	Test accuracy	activity classes predicted

Verification:

- Evaluated on 2,947 held-out test samples.
- Random Forest with 250 trees.
- Feature extraction consistency verified by the training pipeline.

Where It Struggles

	Training data	Live browser
Placement	Phone at the waist	Hand, pocket or table
Sample rate	Steady 50 Hz	About 60 Hz, varies
Axes and gravity	One phone model	Differs by phone and OS
Body acceleration	Always provided	May be missing on some Android phones

What we do about it

- Convert units to match training
- Vote over the last 5 windows
- Keep the phone steady in the demo
- Sitting and standing can look alike

Conclusions, Limitations & Future Work

Conclusions

- Working pipeline from dataset to live web app
- Same features in training and live use
- Light model, real-time response

Limitations

- Waist-worn training data
- Phone and OS differences
- Only 30 volunteers

Future work

- Resample live data to 50 Hz
- Collect our own phone data
- Try a 1D-CNN or LSTM

Thank you